



Guidelines

Midodrine and Perioperative Hypotension

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Allied Health, Pharmacy

Other causes of perioperative hypotension must be excluded before initiating midodrine (e.g. bleeding, hypovolaemia)

Introduction

This guideline provides guidance to all Sir Charles Gairdner Hospital (SCGH) staff involved in the prescription, administration, supply and management of patients on midodrine for perioperative hypotension. This document supports and complies with related legislation, policy and other related documents [here](#).

1.0 General information

Midodrine is a useful treatment for the management of perioperative hypotension. Its availability can prevent the use of excessive IV fluids to treat perioperative hypotension which can delay patient recovery. It can also prevent the need for admission to critical care areas for management with IV vasopressors.

The mechanism of action of midodrine is via α -1 adrenoceptor-mediated vasoconstriction.

2.0 Presentation

- Midodrine 2.5mg and 5mg tablets

3.0 Indications

- Midodrine is approved for use for the management of perioperative hypotension.
- Refer to the [WA Statewide Medicines Formulary \(SMF\)](https://formulary.hdwa.health.wa.gov.au/Home) for full details of restriction (<https://formulary.hdwa.health.wa.gov.au/Home>)

4.0 Contraindications^{1,2}

- Urinary retention
- Persistent and excessive supine hypertension
- Pheochromocytoma
- Severe organic heart disease
- Thyrotoxicosis

5.0 Precautions^{1,2,3}

- Cardiovascular:

- May cause bradycardia due to the vagal reflex, use with caution with concomitant administration of medications that lower heart rate (e.g. digoxin, beta blockers)
- Use with caution in patients with orthostatic hypotension with supine hypertension due to a marked elevation in supine blood pressure
- Use with caution in patients with severe heart disease (e.g. bradycardia, MI, heart failure, cardiac conduction abnormalities) or cardiovascular occlusive disease
- Hepatic
 - Use with caution in hepatic impairment
- Renal
 - Use with caution in renal impairment (CrCl <30mL/min)
- Ophthalmic
 - Patients with history of vision disorders and receiving fludrocortisone are at risk of elevated intraocular pressure
- Prostate disorders
 - Caution is advised due to the risk of urinary retention

6.0 Drug Interactions²

- The risk of hypertension increases with concomitant administration of drugs that increase blood pressure (phenylephrine, pseudoephedrine, ephedrine or thyroid hormones). Avoid concomitant use of drugs that increase blood pressure. If concomitant use cannot be avoided, monitor blood pressure closely.
- Cardiac glycosides (i.e. digoxin) may enhance or precipitate bradycardia, A.V. block or arrhythmia
- Avoid use of MAO inhibitors or linezolid with midodrine
- Midodrine has been used in patients concomitantly treated with salt-retaining steroid therapy (i.e. fludrocortisone acetate), with or without salt supplementation. The potential for supine hypertension should be carefully monitored in these patients and may be minimised by either reducing the dose of fludrocortisone acetate or decreasing the salt intake prior to initiation of treatment with midodrine. Alpha-adrenergic blocking agents, such as prazosin can antagonize the effects of midodrine.

7.0 Dosage

- An initial **loading dose of 20mg** either on the ward prior to transfer to theatre or postoperatively in the Post Anaesthetic Care Unit (PACU)
 - This may be withheld at the discretion of the Anaesthetist.
 - Prescribe as a stat dose on the front of the WA Hospital Medication Chart (WAHMC)
 - Midodrine is to be withheld if the systolic blood pressure (SBP) is greater than 160mmHg (i.e. SBP >160mmHg) at the scheduled time of administration
- Follow with **10mg every 4 hours** postoperatively
 - Prescribe on the "Regular Medicines" section on the WAHMC.
 - Midodrine is to be withheld if the SBP is greater than 130mmHg (i.e. SBP >130mmHg) at the scheduled time of administration
- Regular dosing may begin no earlier than 2 hours after the initial 20mg loading dose. Due to pharmacokinetics, the first regular dose on the WAHMC should ideally be 3-4 hours after initial loading dose.
- A maximum of **70mg midodrine is permitted in the first 24 hours post-op** (inclusive of the initial 20mg loading dose)

- Cease prescription 24 to 72 hours post-operatively and any derangements to the patient's blood pressure outside of normal parameters should warrant review as directed by the Observation and Response Chart (ORC) (i.e. senior nurse / medical review / MET call) as appropriate.

8.0 Administrations

- Midodrine tablets should be administered orally with water
 - Tablets can be crushed if required for patients with swallowing difficulties
- No restriction with fasting status

9.0 Monitoring

- Expect a gradual desired effect within an hour and can last for up to 6 hours
- SBP as high as 200mmHg can occur with midodrine (nursing staff to follow prompts for escalation in ORC)
 - Medical review may be indicated
- On completion of post-operative observations, continue 4-hourly core physiological observations as a minimum for the duration of the midodrine treatment
 - If SBP is outside the normal range, monitoring should proceed as directed on the ORC

10.0 Nursing Management

It is the responsibility of the Registered Nurse when dispensing the medication to ensure:

- Safe administration of the drug in accordance with Nursing Practice Guidelines as identified in key documents
- Correct documentation on the medication chart (and PACU record when patient in PACU)
- Withhold midodrine if systolic blood pressure is greater than 130mmHg (i.e. SBP >130mmHg) at the time the medication is due to be dispensed
 - Mark "withheld" in the box for that dosing time

11.0 Adverse Effects¹

- Hypertension, bradycardia, dizziness, piloerection, pruritus, shivering, paraesthesia, dysuria, urinary retention and/or frequency

12.0 Pregnancy and Breastfeeding⁴

- TGA Category C: use with caution
- Use with caution in breast feeding, insufficient data
- Contact Obstetric Medicines Information Services phone: 08 6458 2723 for more information.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- NSQHS Standards: 1,4,6,8

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline No.51 Medication Management](#)
- [SCGOPCG Hip Fracture Management Guideline](#)

Reference

1. Australian Medicines Handbook [Internet]. Adelaide (SA): Australian Medicines Handbook Pty Ltd; Midodrine [cited 2023 Aug 15]. Available from: Australian Medicines Handbook.
2. MIMS Online [Internet]. Crows Nest (NSW): MIMS Australia Pty Ltd; 2023. Midodrine Vasodrine; [updated 2022 June 28; cited 2023 Aug 15]. Available from: MIMS Online
3. The Renal Drug Database [Internet]. Oxon (UK): CRC Press;2023. Midodrine [cited 2023 Aug15]. Available from The Renal Drug Database.
4. [Women and Newborn Health Service Midodrine Guideline](#)

Authorisation & Version Control

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Standards Applicable	NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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